



UNIVERSITY OF UYO · 300 LEVEL PROJECT

campusfind

Lost and found for the whole campus, with automatic match suggestions.

47

seeded posts

40

students

17

claims

Contents

1. Overview
2. How the core logic works
3. Architecture and data model
4. Screen walkthrough
5. Mobile view
6. Running locally
7. Testing
8. Deployment
9. Five minute presentation script

Demo login: student@uniuyo.edu.ng / student123 (Ini Ekpo). Screens captured with the clock frozen at 2026-10-05.

1. Overview

Lost items on campus are usually announced in a few WhatsApp groups and then forgotten. Finders do not know who to give things to, and owners do not know where to look. campusfind gives every lost or found item one post with a photo, a category, a place and a date, and then suggests posts that look like the same item.

Feature	Details
Post an item	Lost or found, title, category, place (with suggestions), date, details and an optional photo.
Search and filter	Free text search, Lost/Found toggle, 10 categories, Active or Returned.
Possible matches	Every post lists open posts of the opposite type that are probably the same item, with a score and reasons.
Contact the poster	Email, phone and WhatsApp links on every post.
Claims	Send a claim with a message only the owner could write. The poster accepts or declines.
Mark as claimed and resolved	Accepting a claim marks the item Claimed and declines the others. Marking it returned closes the post.
Privacy	Claims are only visible to the poster. My posts and My claims only show the signed-in student's own rows.

Design

The interface follows the visual style of Grok: a black canvas, white text, thin grey borders, rounded pill controls, a large prompt-style search box and monospace labels. It is a normal web app, not a chatbot.

2. How the core logic works

The rules are pure functions in `src/lib/matching.ts` and `src/lib/items.ts`. They take "today" as an argument. Set `CAMPUSFIND_TODAY=YYYY-MM-DD` to freeze the clock for demos and tests.

Possible match score (0 to 100)

Signal	Points	Notes
Hard filters	score 0	Same type (two lost posts), same poster, a resolved post, or found more than 1 day before it was lost or more than 45 days after.
Same category	+35	Different categories only match if they share at least 2 words.
Shared words	up to +40	Word overlap of title and details (Jaccard x 100) plus 8 per shared title word.
Same place	+15 (or +7 nearby)	Nearby means a shared place word, ignoring generic words like Main, Campus or Faculty.
Time gap	+10 within 3 days, +5 within 14	Measured from the lost date to the found date.

Only scores of 45 or more are shown, best first, top 5. Labels: 80+ Strong, 62+ Likely, otherwise Possible.

Words

`tokenize` lowercases the text, drops punctuation, numbers and common words (my, lost, near, the), removes plural endings (keys to key, batteries to battery) and maps brands and synonyms to one word: iPhone, Tecno and Samsung become phone; AirPods and Oraimo become earbud; backpack and handbag become bag; specs and spectacles become glass. So "Black Oraimo earbuds" matches "earbuds charging case".

Claim rules

```
claimBlocker(item, user, alreadyClaimed)
  own post      -> "This is your own post"
  resolved      -> "This item has been returned to its owner"
  claimed       -> "The poster has already accepted a claim"
  already sent  -> "You have already sent a message about this item"
  otherwise     -> allowed
```

`decideClaim` runs in a transaction and locks the item row. Accepting one claim sets it to accepted, declines the other pending claims and sets the item to **claimed**. Only the poster can accept, decline or mark the item **resolved**. A unique constraint on (item_id, claimant_id) stops duplicate claims.

Other rules

Dates cannot be in the future or older than 6 months. Open posts older than 30 days get an "Old post" flag. Phone numbers like 0803 123 4567 become WhatsApp links with the +234 prefix.

3. Architecture and data model

Browser

Server-rendered pages. Client components only for the post form (photo preview), claim form and live refresh.

Next.js 16 App Router

Server Components read, Server Actions write. proxy.ts redirects visitors without a session. /api/health pings Postgres.

PostgreSQL

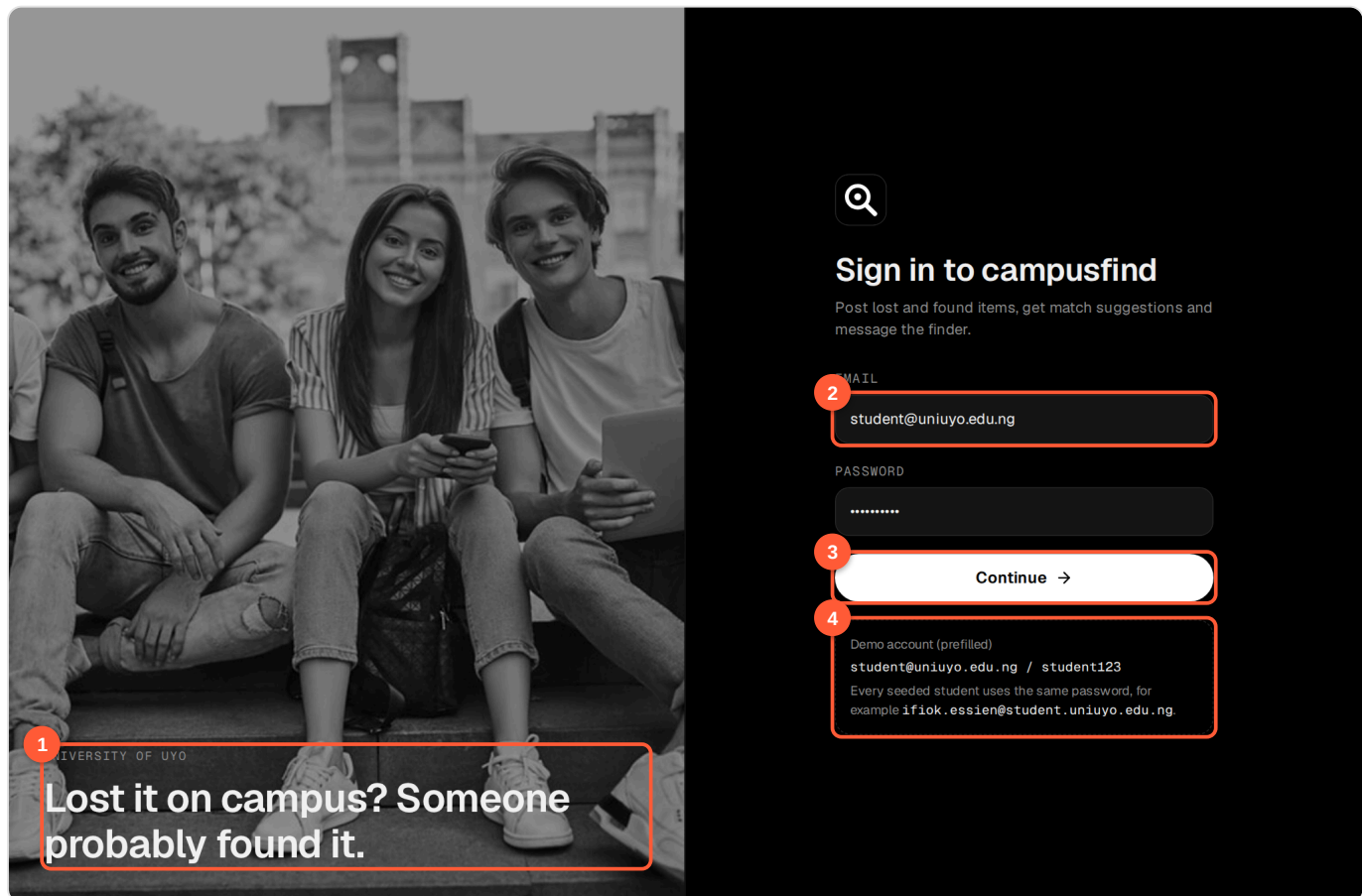
Drizzle ORM schema and SQL migrations from drizzle-kit. Uploaded photos are stored as bytea.

Table	Columns	Rules
users	id, name, email, phone, password_hash, role, department, created_at	email unique
items	id, type (lost, found), title, description, category, location, happened_on, image_url, status (open, claimed, resolved), posted_by, created_at, resolved_at	indexes on type+status, category, posted_by
claims	id, item_id, claimant_id, message, contact, status (pending, accepted, declined), created_at	unique (item_id, claimant_id), message at least 10 characters, cascades
uploads	id (uuid), mime, data (bytea), uploaded_by, created_at	served by /api/images/[id]

Route	Purpose
/login	Sign in (demo prefilled)
/	Feed with search and filters
/post	Post a lost or found item
/items/[id]	Item, contact, claim form or owner panel, possible matches
/mine, /claims	The signed-in student's posts and claims
/api/health, /api/images/[id]	Health check, uploaded photos

4. Screen walkthrough

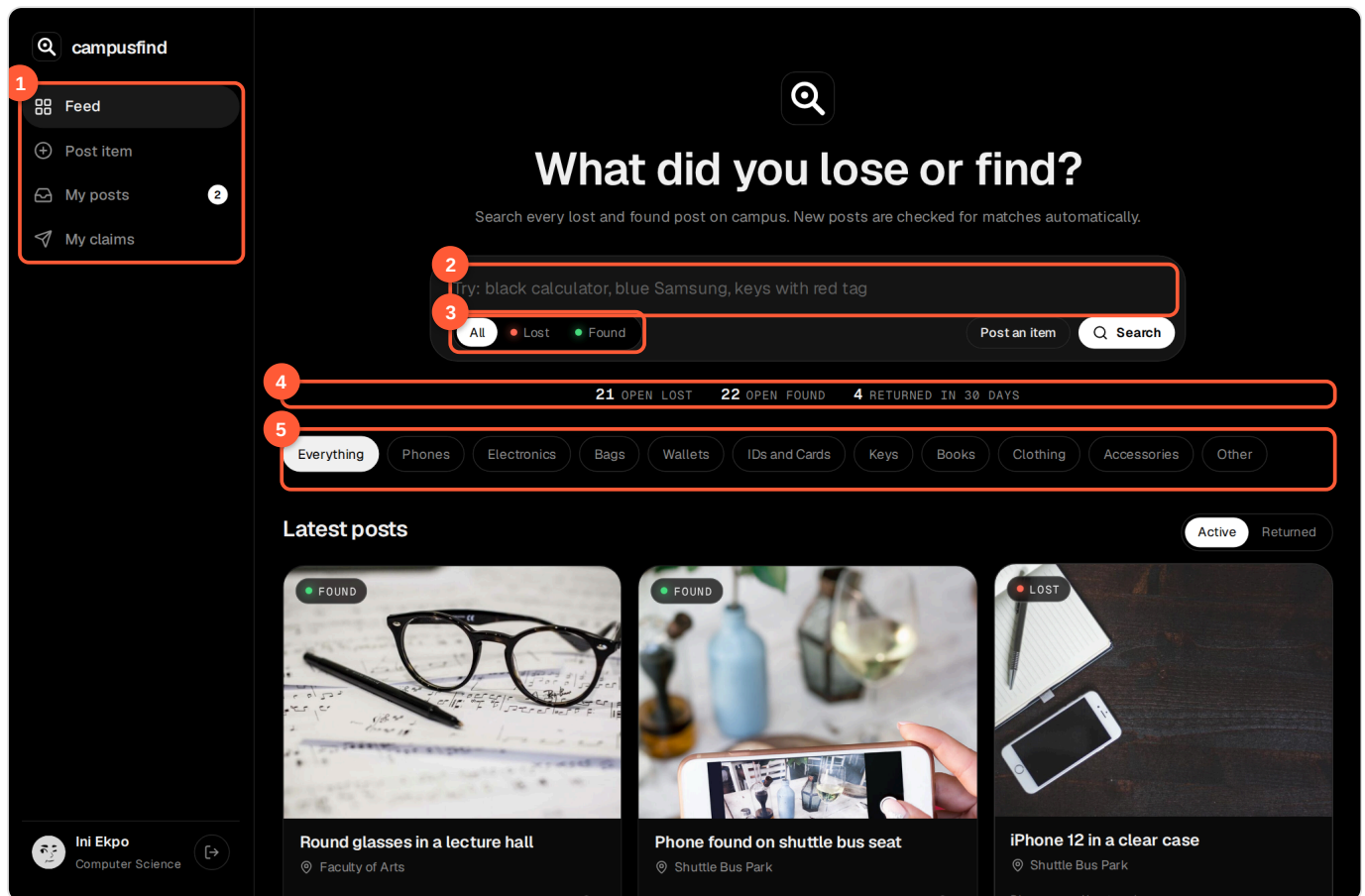
Sign in



Sign-in page.

- 1 Brand panel with a real campus photo (Creative Commons), shown in greyscale to fit the black theme.
- 2 Email and password are prefilled with the demo student, Ini Ekpo.
- 3 Continue posts to a Server Action, checks the bcrypt hash and sets a signed JWT in an HTTP-only cookie.
- 4 Every seeded student uses the same password, which makes it easy to sign in as the other side of a claim.

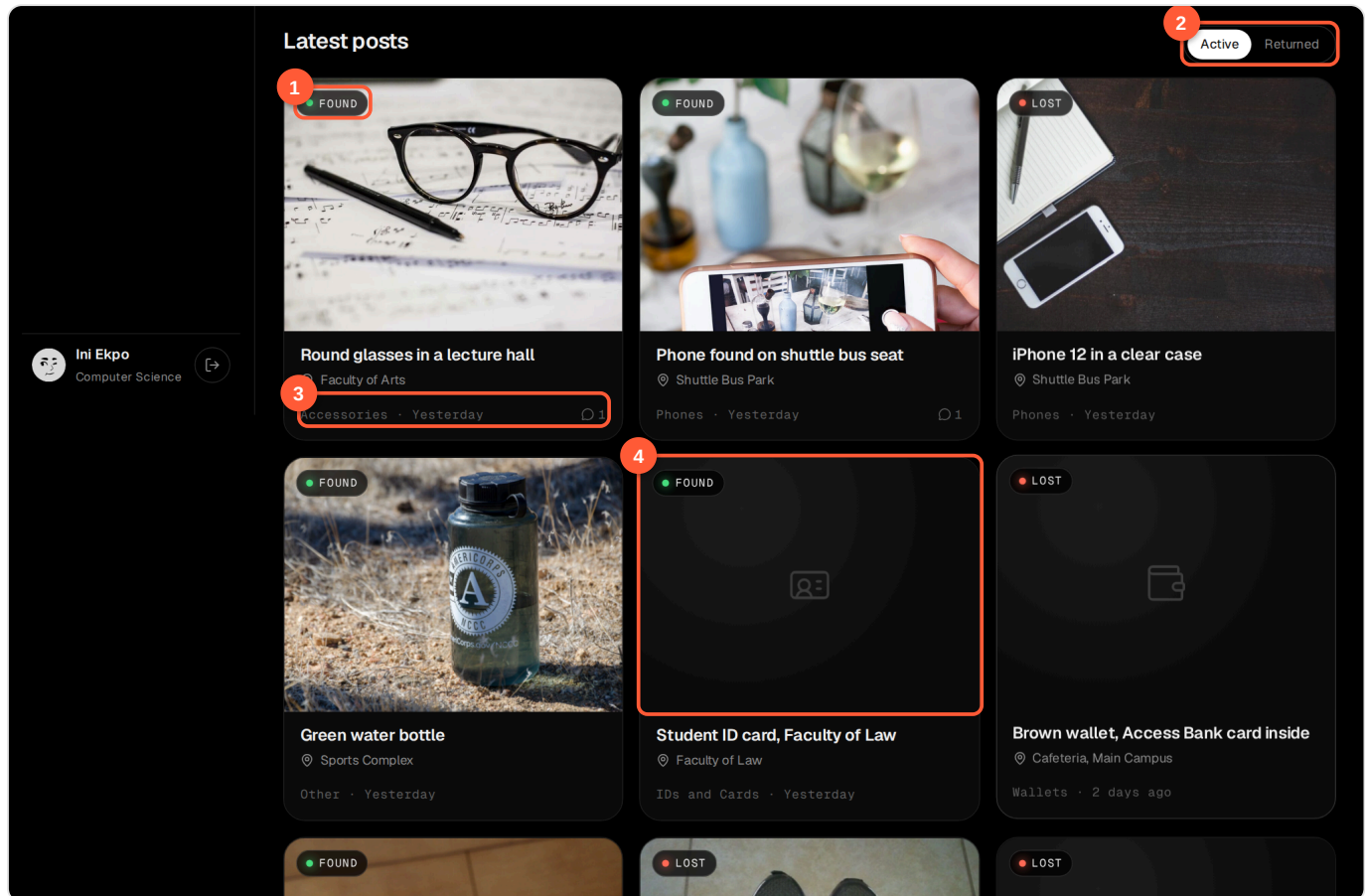
Feed



Top of the feed.

- 1 Sidebar: Feed, Post item, My posts and My claims. The white badge counts new claims waiting on your posts.
- 2 One search box for titles, details and places.
- 3 All, Lost or Found toggle.
- 4 Live totals: open lost, open found and items returned in the last 30 days.
- 5 Category filters. Filters live in the URL.

Item cards



Latest posts.

- 1 LOST or FOUND tag with a red or green dot.
- 2 Active posts or items already returned to their owners.
- 3 Category, how long ago, and how many people have responded.
- 4 Posts without a photo get a category icon instead.

Post form with errors

The screenshot shows a web form titled "Post a lost or found item" on a dark-themed interface. The form is for creating a new post and includes several input fields. Five red circles with numbers 1 through 5 point to specific error messages:

- 1** Points to a red error message: "Some fields need attention."
- 2** Points to the "LOST/FOUND" section, which has two radio buttons: "Lost something" (selected) and "Found something".
- 3** Points to the "TITLE" input field, which contains the text "e.g. Black Casio calculator". Below the field is a red error message: "Give it a short title, at least 4 characters".
- 4** Points to the "DATE" input field, which contains the date "10/05/2026".
- 5** Points to the "PHOTO" section, which includes a photo picker icon and the text "Add a photo". Below this is a red error message: "Add a few details, at least 15 characters".

The form also includes a sidebar with navigation links: "Feed", "Post item", "My posts", and "My claims". The user's profile "Ini Ekpo" is visible at the bottom left.

Submitting the post form empty.

- 1 Summary when something is missing.
- 2 Lost or Found choice. Zod reports 'Choose lost or found' inline when it is skipped.
- 3 Each field shows its own error right under the input.
- 4 Date defaults to today. Future dates and dates older than 6 months are rejected.
- 5 Photo picker with a live preview. Uploads are checked for type and size and stored in Postgres.

After posting: possible matches

campusfind

Feed

Post item

My posts2

My claims

← Feed

1

2

3

4

5

Posted. We found 4 possible matches below.

FOUND



#0048

Black scientific calculator

FOUND ON

Mon, 5 Oct

Today

WHERE

Faculty of Science

CATEGORY

Electronics

Found a black Casio calculator under a seat in Lecture Theatre 2 after the test.

Posted by

 You

Computer Science · 5 Oct, 1:37 am

Messages on your post0

No one has responded yet.

Mark as returned to owner

Possible matches

Lost posts with the same category and similar words, scored by our matcher.

LOST

Faculty of Science · 2 days ago



Casio fx-991ES calculator

Same category: Electronics

Shared words: black, scientific, calculator, lecture, theatre

Same place: Faculty of Science

Found 2 days after it was lost

STRONG MATCH

100%

LOST

Faculty of Engineering · 6 days ago



HP scientific calculator

Same category: Electronics

Shared words: black, scientific, calculator

LIKELY MATCH

73%

LOST

Students' Union Building · 1 week ago



White over-ear headphones

Same category: Electronics

Shared words: after

POSSIBLE MATCH

46%

LOST

ICT Centre · 3 days ago



Red USB-C charging cable

POSSIBLE MATCH

45%

Ini Ekpo

Computer Science

A found calculator posted with a photo. The matching lost calculator is first.

- 1 Straight after posting, the page says how many possible matches were found.
- 2 The uploaded photo, served from /api/images/[id].
- 3 Top match: the lost Casio fx-991ES. Same category, shared words, same place, lost the day before.
- 4 Score out of 100 with a label (Strong, Likely or Possible).
- 5 Reasons are listed so students can see why a post was suggested.

campusfind documentation

Page 9 of 19

Claiming a found item

The screenshot displays the campusfind app interface. On the left is a dark sidebar with a search bar containing 'campusfind', a 'Feed' button, and navigation options: 'Post item', 'My posts' (with a '2' badge), and 'My claims'. The main content area shows a post titled 'Earbuds charging case found in library reading room' with a '#0002' tag. The post features a photo of a black 'FAIRPHONE' charging case and two black earbuds. Below the photo, a table provides details: 'FOUND ON' (Sat, 3 Oct, 2 days ago), 'WHERE' (University Library), and 'CATEGORY' (Electronics). The post text describes finding a black earbuds charging case with both earbuds inside on a table in the ground floor reading room. The poster is 'Idara Umoh' from 'Pharmacy', dated '3 Oct, 4:37 am'. A red box labeled '1' highlights the contact options: 'Email', '08168234508', and 'WhatsApp'. Below this, a section titled 'Is this yours?' contains a text input field labeled 'Describe something only the owner would know: marks, contents, lock screen.', a 'PHONE (OPTIONAL)' field with '0803 123 4567', and a red box labeled '3' around a button that says 'This is mine, send claim'. At the bottom, it says '1 person has responded to this post.' The bottom of the sidebar shows the user 'Ini Ekpo' from 'Computer Science'.

The demo student looking at earbuds someone found.

- 1 Contact the poster by email, phone or WhatsApp (link built from the Nigerian number).
- 2 Claim message: the owner describes something only they would know.
- 3 Sends the claim. A student can claim an item once; their own posts cannot be claimed.
- 4 The demo student's own lost earbuds show up as a strong match.

Owner view

campusfind

Feed

Post item

My posts 2

My claims

FOUND

#0003

Brown leather wallet with ATM card

FOUND ON	WHERE	CATEGORY
Sat, 3 Oct 2 days ago	Cafeteria, Main Campus	Wallets

Brown leather wallet found under a table at the cafeteria around lunch time. It has an ATM card and some cash inside. Tell me the bank on the card and roughly how much cash to claim it.

Posted by

You
Computer Science · 3 Oct, 5:14 am

1 messages on your post 2

2 Ifiok Essien
4 Oct, 8:00 am · 08095234505 PENDING

This is mine. The card is an Access Bank debit card in the name Ifiok Essien and there was about N4,500 in cash, mostly N1,000 notes.

✓ Accept × Decline

3 Precious Etuk
4 Oct, 8:00 am · 09017234527 PENDING

I lost a brown wallet last week, I think this could be mine.

✓ Accept × Decline

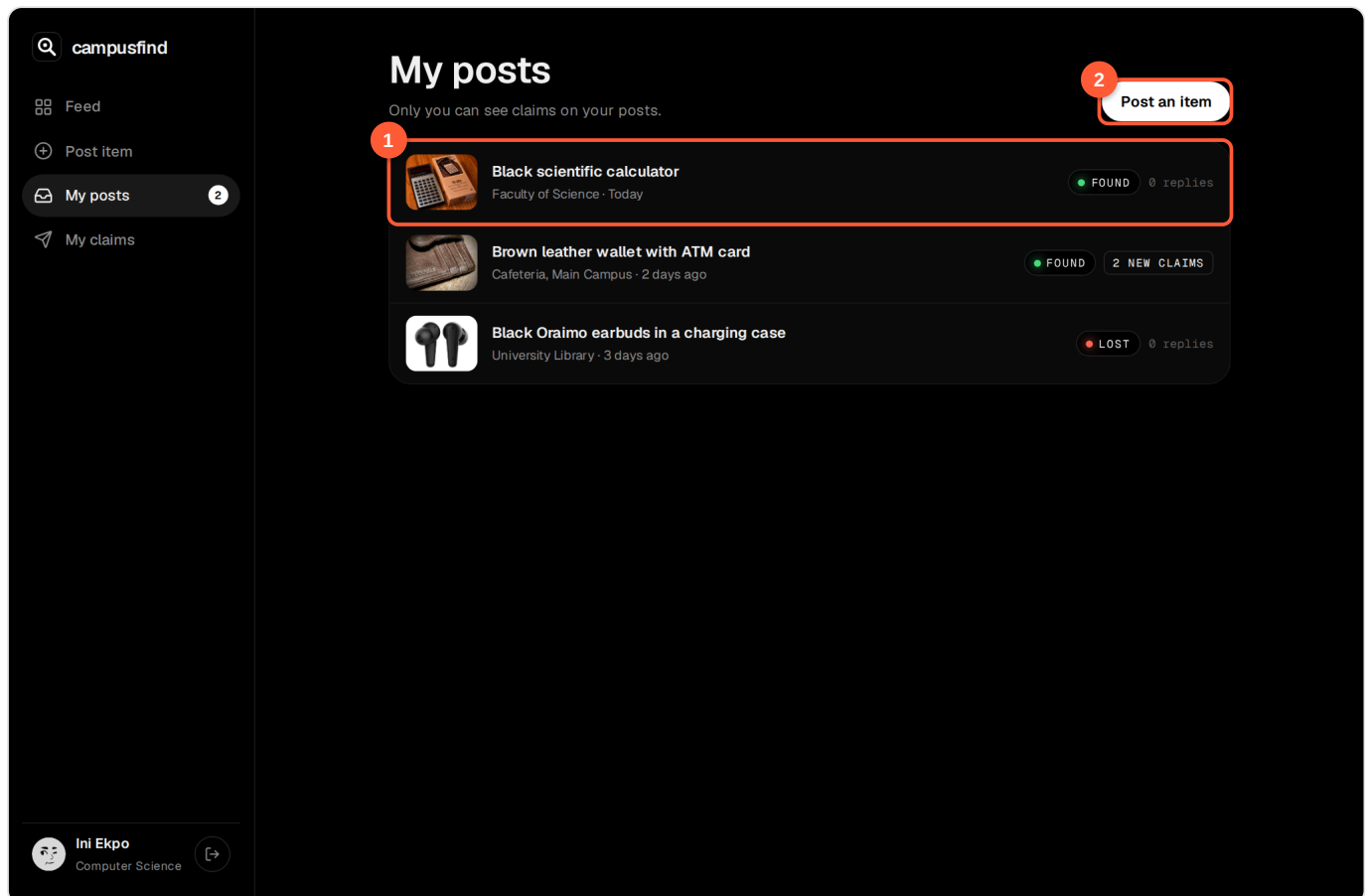
4 Mark as returned to owner

Ini Ekpo
Computer Science

The demo student's own found wallet with two claims.

- 1 Only the poster sees the messages on a post.
- 2 A detailed claim: bank name and amount. Accept or Decline.
- 3 A vague claim that the poster can decline.
- 4 Mark as returned closes the post and moves it to the Returned tab.
- 5 The matching lost wallet, posted by the student who sent the detailed claim.

My posts

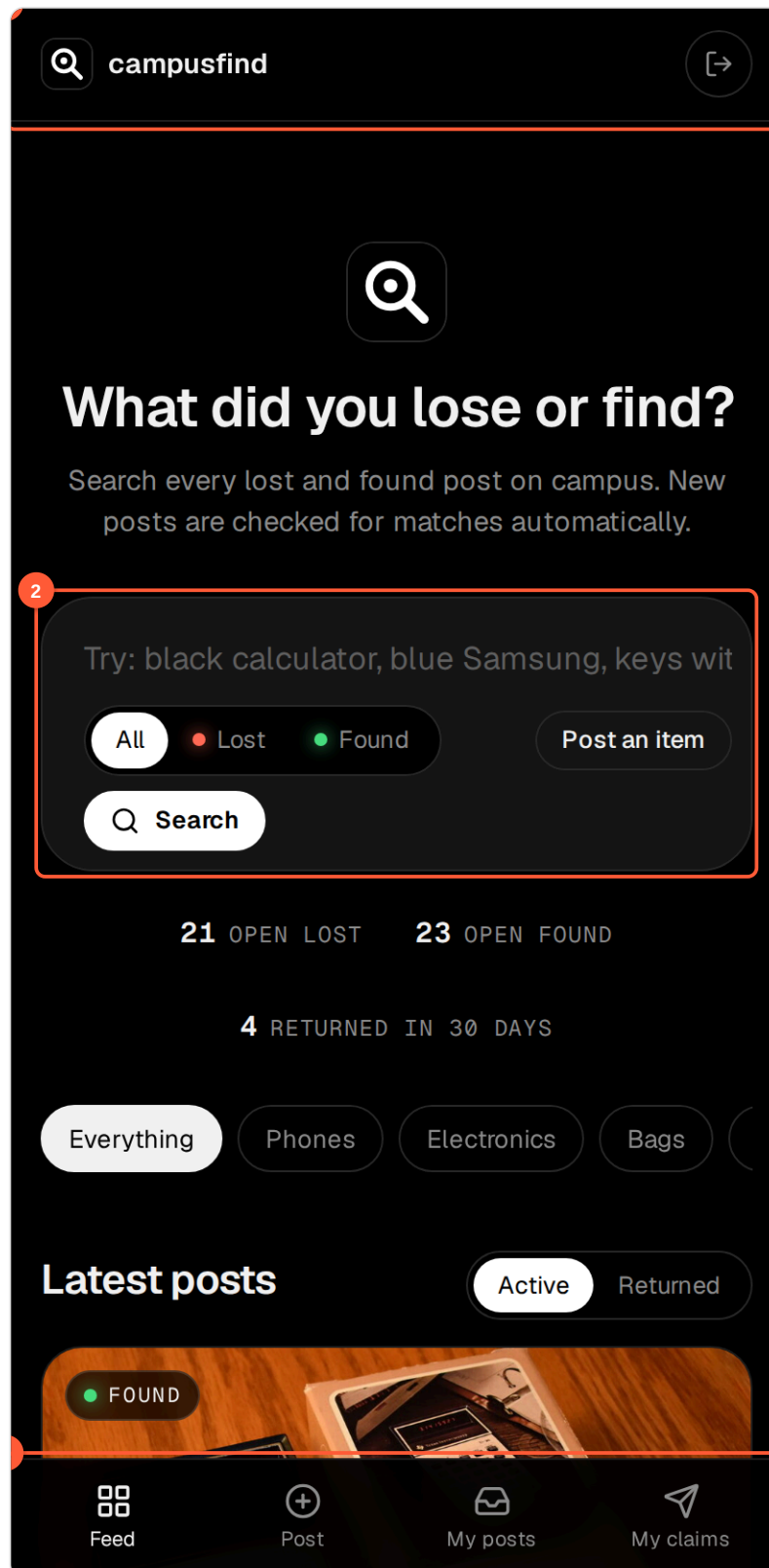


Posts by the signed-in student.

- 1 Your own posts only. New claims are highlighted.
- 2 Shortcut to post another item.

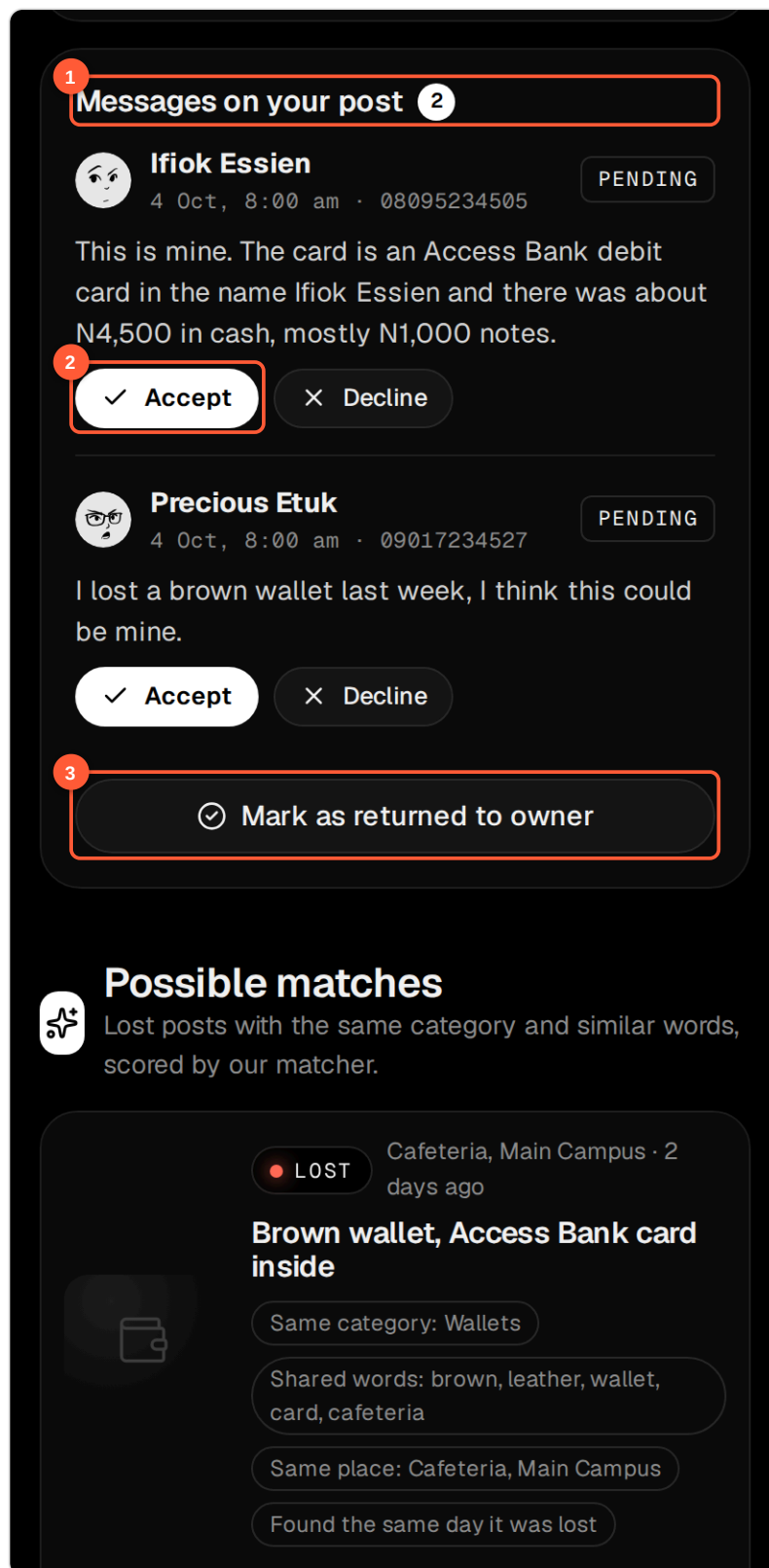
5. Mobile view

Grids use `minmax(0, 1fr)` so nothing pushes the page sideways. The mobile e2e test runs on a Pixel 7 profile and fails on horizontal scroll.



Feed on a phone.

- 1 On phones the sidebar becomes a top bar.
- 2 Search and the Lost or Found toggle stay together.
- 3 Bottom tab bar for the four main screens.



Claims on the wallet post, on a phone.

- 1 Messages on your post, under the photo and details.
- 2 Accept and Decline buttons sized for thumbs.
- 3 Mark as returned, full width.

6. Running locally

```
service postgresql start
sudo -u postgres createdb campusfind
sudo -u postgres createdb campusfind_test
cp .env.example .env      # DATABASE_URL, SESSION_SECRET
npm install
npm run db:migrate
npm run db:seed           # resets and seeds posts dated relative to today
npm run dev               # http://localhost:3000
```

Script	What it does
dev / build / start	Development server, production build, production server
start:prod	Migrate, seed only when empty, next start -H 0.0.0.0 (Railway)
db:generate, db:migrate, db:seed	New migration, apply migrations, reset and seed
test:unit, test:integration, test:e2e	Test suites
docs:pdf	Rebuild this PDF

For the live demo, a sample photo is included at `public/demo/found-calculator.jpg`.

7. Testing

23

unit tests passed

9

integration tests passed

6

e2e tests passed

Suite	Covers
Unit (Vitest)	Tokenizer, stemming and synonyms, match score and its hard filters, symmetry, ranking, claim and resolve rules, date labels and validation, WhatsApp links, Zod schemas, photo checks, JWT sessions.
Integration (Vitest + Postgres)	Seed volume and states, matches for a newly posted calculator, claim once, no claims on own post, claims hidden from non-owners, accept declines others then resolve, unique constraint, scoping of My posts, search and filters.
End to end (Playwright)	Login redirect and prefill, search with type filter, inline errors, posting a found item with a photo and seeing the match, a full claim then accept then resolve between two browser sessions, and a mobile run with no sideways scroll.

Latest run: unit 23, integration 9, 0 failed; e2e 6 of 6 passed (5 desktop, 1 mobile).

8. Deployment

campusfind runs on Railway in the **school-projects** project as the **campus-find** service, with its own **campus-find-postgres** database. The GitHub repository is connected, so each push to main deploys.

Setting	Value
Build	npm run build
Start	npm run start:prod
Health check	/api/health (503 if Postgres is unreachable)
DATABASE_URL	\${{campus-find-postgres.DATABASE_URL}}
SESSION_SECRET	random value
NODE_ENV	production

9. Five minute presentation script

Time	What to do and say
0:00	Problem: lost items are announced in WhatsApp groups and forgotten. campusfind keeps every lost and found item in one searchable place and suggests matches.
0:30	Sign in with the prefilled demo account. Show the search box, the Lost/Found toggle, the totals and category chips. Search "keys".
1:10	Click Post item. Submit it empty to show the inline errors. Then choose Found, title "Black scientific calculator", Electronics, Faculty of Science, a short description, and attach public/demo/found-calculator.jpg.
2:00	Post. The page says it found possible matches. The lost Casio fx-991ES is first with a high score. Read out the reasons: same category, shared words, same place, lost the day before. This is the wow moment.
2:45	Open "Earbuds charging case found in library reading room". Show the email, phone and WhatsApp buttons. Write a claim describing the scratch and send it.
3:30	Open My posts, then the brown leather wallet. Show the two claims: one specific, one vague. Accept the specific one: the item becomes Claimed and the other claim is declined.
4:05	Click Mark as returned. Show it under the Returned tab on the feed.
4:25	Architecture: Next.js Server Components and Server Actions, Drizzle with Postgres, pure matching functions, tests (quote the counts), Railway with a health check.
4:50	Next steps: email alerts when a match appears, and a Security desk account for items handed in. Questions.

Image credits

Item and campus photos are Creative Commons images found through Openverse, resized and stored in `public/images`. Authors, licences and links are in `public/images/items/credits.json` and `public/images/scenes/credits.json`.